

Finally, note that when $s = 0$ we have $\Gamma(1) = \lim_{s \rightarrow 0} s\Gamma(s)$.

An alternate proof of Theorem 1.3, which is interesting in its own right and whose ideas recur later, is obtained by splitting the integral for $\Gamma(s)$ defined on $\operatorname{Re}(s) > 0$ as follows:

$$\Gamma(s) = \int_0^1 e^{-t} t^{s-1} dt + \int_1^\infty e^{-t} t^{s-1} dt.$$

The integral on the far right defines an entire function; also expanding e^{-t} in a power series and integrating term by term gives

$$\int_0^1 e^{-t} t^{s-1} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)}.$$

Therefore

$$(3) \quad \Gamma(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t} t^{s-1} dt \quad \text{for } \operatorname{Re}(s) > 0.$$

Finally, the series defines a meromorphic function on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $s = -n$. To prove this, we argue as follows. For a fixed $R > 0$ we may split the sum in two parts

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+s)} = \sum_{n=0}^N \frac{(-1)^n}{n!(n+s)} + \sum_{n=N+1}^{\infty} \frac{(-1)^n}{n!(n+s)},$$

where N is an integer chosen so that $N > 2R$. The first sum, which is finite, defines a meromorphic function in the disc $|s| < R$ with poles at the desired points and the correct residues. The second sum converges uniformly in that disc, hence defines a holomorphic function there, since $n > N > 2R$ and $|n+s| \geq R$ imply

$$\left| \frac{(-1)^n}{n!(n+s)} \right| \leq \frac{1}{n!R}.$$

Since R was arbitrary, we conclude that the series in (3) has the desired properties.

In particular, the relation (3) now holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

The following identity reveals the symmetry of Γ about the line $\operatorname{Re}(s) = 1/2$.

Theorem 1.4 For all $s \in \mathbb{C}$,

$$(4) \quad \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}.$$

Observe that $\Gamma(1-s)$ has simple poles at the positive integers $s = 1, 2, 3, \dots$, so that $\Gamma(s)\Gamma(1-s)$ is a meromorphic function on $\mathbb{C}$ with simple poles at *all* the integers, a property also shared by $\pi/\sin \pi s$.

To prove the identity, it suffices to do so for $0 < s < 1$ since it then holds on all of $\mathbb{C}$ by analytic continuation.

Lemma 1.5 For $0 < a < 1$, $\int_0^\infty \frac{v^{a-1}}{1+v} dv = \frac{\pi}{\sin \pi a}$.

Proof. We observe first that

$$\int_0^\infty \frac{v^{a-1}}{1+v} dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x} dx,$$

which follows by making the change of variables $v = e^x$. However, using contour integration, we saw in Example 2 of Section 2.1 in Chapter 3, that the second integral equals $\pi/\sin \pi a$, as desired.

To establish the theorem, we first note that for $0 < s < 1$ we may write

$$\Gamma(1-s) = \int_0^\infty e^{-u} u^{-s} du = t \int_0^\infty e^{-vt} (vt)^{-s} dv,$$

where for $t > 0$ we made the change of variables $vt = u$. This trick then gives

$$\begin{aligned} \Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t} t^{s-1} \Gamma(1-s) dt \\ &= \int_0^\infty e^{-t} t^{s-1} \left(t \int_0^\infty e^{-vt} (vt)^{-s} dv \right) dt \\ &= \int_0^\infty \int_0^\infty e^{-t[1+v]} v^{-s} dv dt \\ &= \int_0^\infty \frac{v^{-s}}{1+v} dv \\ &= \frac{\pi}{\sin \pi(1-s)} \\ &= \frac{\pi}{\sin \pi s}, \end{aligned}$$

and the theorem is proved.

In particular, by putting $s = 1/2$, and noting that $\Gamma(s) > 0$ whenever $s > 0$, we find that

$$\Gamma(1/2) = \sqrt{\pi}.$$

We continue our study of the gamma function by considering its reciprocal, which turns out to be an entire function with remarkably simple properties.

Theorem 1.6 *The function Γ has the following properties:*

- (i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s = 0, -1, -2, \dots$ and it vanishes nowhere else.
- (ii) $1/\Gamma(s)$ has growth

$$\left| \frac{1}{\Gamma(s)} \right| \leq c_1 e^{c_2 |s| \log |s|}.$$

Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that

$$\left| \frac{1}{\Gamma(s)} \right| \leq c(\epsilon) e^{c_2 |s|^{1+\epsilon}}.$$

Proof. By the theorem we may write

$$(5) \quad \frac{1}{\Gamma(s)} = \Gamma(1-s) \frac{\sin \pi s}{\pi},$$

so the simple poles of $\Gamma(1-s)$, which are at $s = 1, 2, 3, \dots$ are cancelled by the simple zeros of $\sin \pi s$, and therefore $1/\Gamma$ is entire with simple zeros at $s = 0, -1, -2, -3, \dots$

To prove the estimate, we begin by showing that

$$\int_1^\infty e^{-t} t^\sigma dt \leq e^{(\sigma+1) \log(\sigma+1)}$$

whenever $\sigma = \operatorname{Re}(s)$ is positive. Choose n so that $\sigma \leq n \leq \sigma + 1$. Then

$$\begin{aligned} \int_1^\infty e^{-t} t^\sigma dt &\leq \int_0^\infty e^{-t} t^n dt \\ &= n! \\ &\leq n^n \\ &= e^{n \log n} \\ &\leq e^{(\sigma+1) \log(\sigma+1)}. \end{aligned}$$

Since the relation (3) holds on all of $\mathbb{C}$, we see from (5) that

$$\frac{1}{\Gamma(s)} = \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \right) \frac{\sin \pi s}{\pi} + \left(\int_1^{\infty} e^{-t} t^{-s} dt \right) \frac{\sin \pi s}{\pi}.$$

However, from our previous observation,

$$\left| \int_1^{\infty} e^{-t} t^{-s} dt \right| \leq \int_1^{\infty} e^{-t} t^{|\sigma|} dt \leq e^{(|\sigma|+1) \log(|\sigma|+1)},$$

and because $|\sin \pi s| \leq e^{\pi|s|}$ (by Euler's formula for the sine function) we find that the second term in the formula for $1/\Gamma(s)$ is dominated by $ce^{(|s|+1) \log(|s|+1)} e^{\pi|s|}$, which is itself majorized by $c_1 e^{c_2|s| \log|s|}$. Next, we consider the term

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi}.$$

There are two cases: $|\operatorname{Im}(s)| > 1$ and $|\operatorname{Im}(s)| \leq 1$. In the first case, this expression is dominated in absolute value by $ce^{\pi|s|}$. If $|\operatorname{Im}(s)| \leq 1$, we choose k to be the integer so that $k - 1/2 \leq \operatorname{Re}(s) < k + 1/2$. Then if $k \geq 1$,

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi} &= (-1)^{k-1} \frac{\sin \pi s}{(k-1)!(k-s)\pi} + \\ &+ \sum_{n \neq k-1} (-1)^n \frac{\sin \pi s}{n!(n+1-s)\pi}. \end{aligned}$$

Both terms on the right are bounded; the first because $\sin \pi s$ vanishes at $s = k$, and the second because the sum is majorized by $c \sum 1/n!$.

When $k \leq 0$, then $\operatorname{Re}(s) < 1/2$ by our supposition, and $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1-s)}$ is again bounded by $c \sum 1/n!$. This concludes the proof of the theorem.

The fact that $1/\Gamma$ satisfies the type of growth conditions discussed in Chapter 5 leads naturally to the product formula for the function $1/\Gamma$, which we treat next.

Theorem 1.7 *For all $s \in \mathbb{C}$,*

$$\frac{1}{\Gamma(s)} = e^{\gamma s} s \prod_{n=1}^{\infty} \left(1 + \frac{s}{n} \right) e^{-s/n}.$$

The real number γ , which is known as **Euler's constant**, is defined by

$$\gamma = \lim_{N \rightarrow \infty} \sum_{n=1}^N \frac{1}{n} - \log N.$$

The existence of the limit was already proved in Proposition 3.10, Chapter 8 of Book I, but we shall repeat the argument here for completeness. Observe that

$$\sum_{n=1}^N \frac{1}{n} - \log N = \sum_{n=1}^N \frac{1}{n} - \int_1^N \frac{1}{x} dx = \sum_{n=1}^{N-1} \int_n^{n+1} \left[\frac{1}{n} - \frac{1}{x} \right] dx + \frac{1}{N},$$

and by the mean value theorem applied to $f(x) = 1/x$ we have

$$\left| \frac{1}{n} - \frac{1}{x} \right| \leq \frac{1}{n^2} \quad \text{for all } n \leq x \leq n+1.$$

Hence

$$\sum_{n=1}^{\infty} \frac{1}{n} - \log N = \sum_{n=1}^{N-1} a_n + \frac{1}{N}$$

where $|a_n| \leq 1/n^2$. Therefore $\sum a_n$ converges, which proves that the limit defining γ exists. We may now proceed with the proof of the factorization of $1/\Gamma$.

Proof. By the Hadamard factorization theorem and the fact that $1/\Gamma$ is entire, of growth order 1, and has simple zeros at $s = 0, -1, -2, \dots$, we can expand $1/\Gamma$ in a Weierstrass product of the form

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n=1}^{\infty} \left(1 + \frac{s}{n} \right) e^{-s/n}.$$

Here A and B are two constants that are to be determined. Remembering that $s\Gamma(s) \rightarrow 1$ as $s \rightarrow 0$, we find that $B = 0$ (or some integer multiple of $2\pi i$, which of course gives the same result). Putting $s = 1$, and using

the fact that $\Gamma(1) = 1$ yields

$$\begin{aligned}
 e^{-A} &= \prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(1 + \frac{1}{n}\right) e^{-1/n} \\
 &= \lim_{N \rightarrow \infty} e^{\sum_{n=1}^N [\log(1+1/n) - 1/n]} \\
 &= \lim_{N \rightarrow \infty} e^{-(\sum_{n=1}^N 1/n) + \log N + \log(1+1/N)} \\
 &= e^{-\gamma}.
 \end{aligned}$$

Therefore $A = \gamma + 2\pi i k$ for some integer k . Since $\Gamma(s)$ is real whenever s is real, we must have $k = 0$, and the argument is complete.

Note that the proof shows that the function $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function that has:

- (i) simple zeros at $s = 0, -1, -2, \dots$ and vanishes nowhere else, and
- (ii) order of growth ≤ 1 .

Observe that $\sin \pi s$ has a similar characterization (except the zeros are now at *all* the integers). However, while $\sin \pi s$ has a stricter growth estimate of the form $\sin \pi s = O(e^{c|s|})$, this estimate (without the logarithm in the exponent) does not hold for $1/\Gamma(s)$ as Exercise 12 demonstrates.

2 The zeta function

The Riemann **zeta function** is initially defined for real $s > 1$ by the convergent series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

As in the case of the gamma function, ζ can be continued into the complex plane. There are several proofs of this fact, and we present in the next section the one that relies on the functional equation of ζ .

2.1 Functional equation and analytic continuation

In parallel to the gamma function, we first provide a simple extension of ζ to a half-plane in $\mathbb{C}$.